

STUDENT PROJECT · TEAM NO. 52

FLIGHT PRICE PREDICTION USING LINEAR REGRESSION

DOCUMENTATION

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PRESENTATION

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VIDEO

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Project link: github.com/adityapatil1807/Flight-Price-Predictions-

02 · PROBLEM FORMULATION

THE PROBLEM AND DATASET

Goal

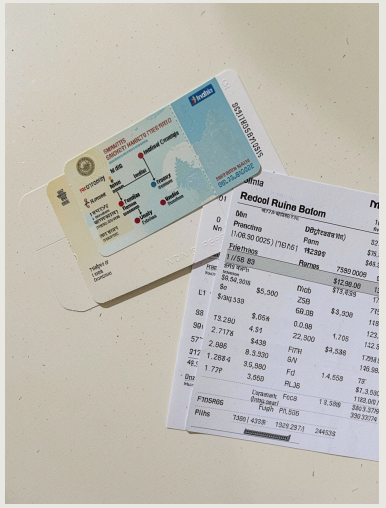
Predict a flight ticket price from route, timing, class, and booking lead time.

Dataset source

Kaggle, with flight-ticket data between six Indian cities.

Target

price



Six Indian cities · flight-ticket data · price prediction

MAIN COLUMNS

airline · flight · source_city · departure_time · stops · arrival_time · destination_city · class · duration · days_left · price

SELECTED INPUTS

source_city · destination_city · departure_time · class · days_left

MODEL ARCHITECTURE & TECHNOLOGIES USED



Input data becomes a repeatable prediction pipeline.

1

Read

Read and understand the dataset.

2

Explore

Explore price patterns with a histogram, a days-left scatter plot, and average-price comparisons by departure time.

3

Prepare

Prepare the data by removing duplicates, handling missing values, and one-hot encoding categorical features.

4

Split and train

Split the data into 80% training and 20% testing, then train a Linear Regression model.

5

Evaluate

Evaluate with MAE, MSE, RMSE, and R^2, then predict a user flight price from route, time, class, and booking days.

LIBRARIES

pandas · NumPy · Matplotlib · scikit-learn

04 · TEST SET EVALUATION

MODEL PERFORMANCE & EVALUATION

Do not invent evaluation values; retain visible fill-in fields until the notebook has been executed.

MAE	MSE	RMSE	R^2
fill after running	fill after running	fill after running	fill after running

FINAL PREDICTION

User flight price

Route, departure time, class, and booking days

VALUE TO BE ENTERED AFTER RUNNING THE NOTEBOOK

ERROR REVIEW & GRAPHS

- Identify the five largest prediction errors.
- Predicted price versus booking days.
- Predicted price versus departure time.

PROJECT LEARNINGS & TEAM CONTRIBUTIONS

TEAM CONTRIBUTIONS

Role	Contributor
Documentation	Aditya Patil
Presentation	Nipun Singhal
Execution lead	Shaurya Kamboj
Video	Jarmik

WHAT WE LEARNED

- Categorical flight details must be converted into model-ready values.
- The model should be tested on data it did not train on.

WHAT WORKED

One preprocessing-and-regression pipeline made prediction organized and repeatable.

AI disclosure: ChatGPT and Claude supported explanations, documentation, and presentation wording; they did not train the model or produce project results.